

Machine learning

Date

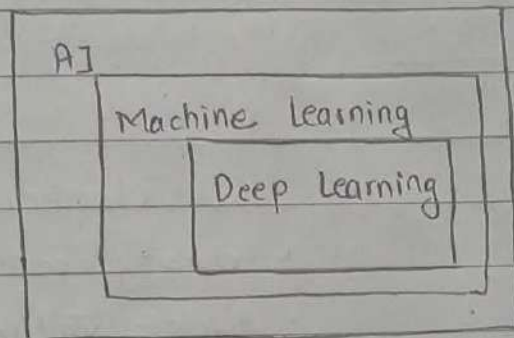
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Generative AI

- technology that can produce various types of content including text, images, audio, synthetic data.

AI

- branch of CS that deals with creation of intelligent data & are system that can learn, reason & act autonomously.

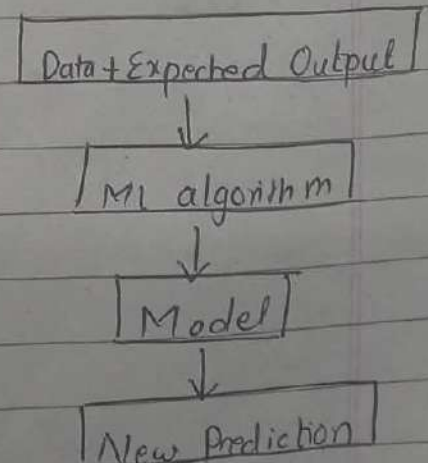


ML

- subfield of AI
- trains models from input data.
- Trained models can make useful predictions from new data drawn from same one used to train model.
- gives ability to learn without explicit programming.

Classes of ML

- Supervised
- Unsupervised
- Semi supervised → Reinforcement learning



Supervised learning

- ↳ model learns from labelled data.
- ↳ comes with tag like name, type or number
- ↳ eg labelled 'spam' or 'not spam' emails
- ↳ eg. # study hrs $\rightarrow$ exam score.

Supervised learning

- ↳ Regression
- ↳ Classification

* Regression

- ↳ predict, numerical value
- ↳ continuous value
- ↳ Best line fit identify
- ↳ Linear regression
 - ↳ simple
 - ↳ multiple
 - ↳ polynomial reg
 - ↳ logistic
 - ↳ KNN

Classification

- ↳ classify, predict category / class
- ↳ discrete value.

↳ KNN

↳ SVM

↳ Naive Bayes

↳ Decision Tree

↳ eg. pass/fail, spam/not spam,
cat/dog, fraud/normal

↳ types

↳ Binary

↳ Multiclass

↳ Multilabel

↳

Unsupervised learning

- discover hidden patterns or structures in the data.
- eg. customer data → Unsupervised learning → Customer Groups
- algorithm discovers group itself.

Major topics

- Clustering
- Dimensionality Reduction
- Anomaly Detection
- Association Rules

Important algorithms

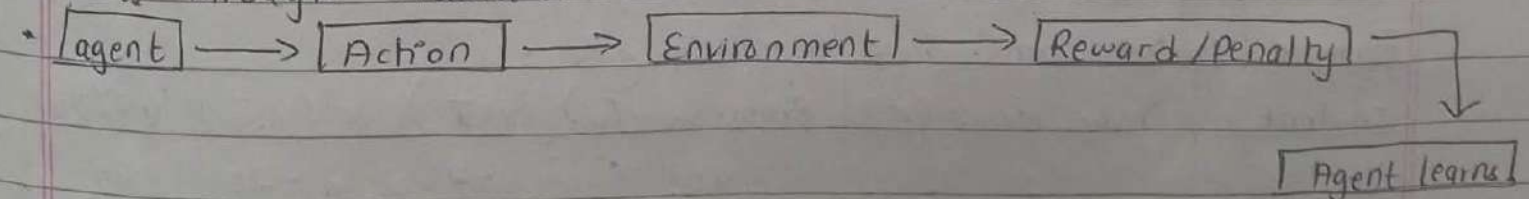
- K-Means
- Hierarchical Clustering
- DBSCAN
- PCA

Semi-supervised learning

- combination of small amount of labelled data and large amount of unlabelled data.
- example
 - 10,000 images
 - 1000 labelled . semi supervised learning use both labeled & unlabelled data.

Reinforcement learning

- learns through interaction with an environment



Linear Regression

Supervised machine learning algorithm used to predict a continuous numerical value.

Input $\rightarrow$ Model $\rightarrow$ Prediction

Eg Predicting exam score based on attendance, study-hrs & internet-usage.

Basic Formula

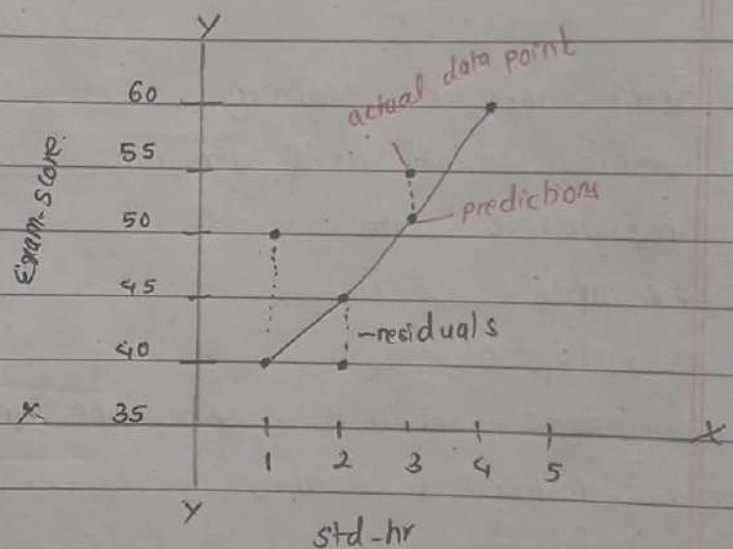
$$\hat{y} = b_0 + b_1 x$$

b_0 = intercept x = input feature

b_1 = slope / coefficient $\hat{y}$ = predicted value

Example.

Study-hr (x)	Actual Exam-score (y)
1	50
2	40
3	55
4	60



where

x (Feature / input)

y = target output

Example: Predict Exam Score from Study hours.

Student	Study-Hours (x)	Exam-Score (y)	$(x - \bar{x})$	$(y - \bar{y})$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$
A	1	40	-2	-20	40	4
B	2	50	-1	-10	10	1
C	3	60	0	0	0	0
D	4	70	1	10	10	1
E	5	80	2	20	40	4
					100	10

If a new student studied 6 hours, what exam score should we predict?

↳ Regression because we are predicting a number.

Given

X = study hours

$\hat{y}$ = predicted score

Y = Exam Score

b_0 = intercept

Now,

b_1 = coefficient / slope

$$\hat{y} = b_0 + b_1 x \longrightarrow$$

n = feature (study hours)

$b_0 = ?$

$b_1 = ?$

Using, Simple Linear Regression (b_1) =
$$\frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$\bar{x} = \frac{1+2+3+4+5}{5} = \frac{15}{5} = 3 \quad = \quad \frac{100}{10} = 10$$

↳ for every additional

$$\bar{y} = \frac{40+50+60+70+80}{5} = \frac{300}{5} = 60$$

1 hr of study, the

predicted exam score will increase by 10.

Now,

$$\begin{aligned} b_0 &= \bar{y} - b_1 \bar{x} \\ &= 60 - 10 \times 3 \\ &= 30 \end{aligned}$$

Our equation is $\boxed{\hat{y} = 30 + 10x}$

So, if a new student studies 6 hours then,

$$\hat{y} = 30 + 10x$$

$$= 30 + 10 \times 6$$

$$= 90 - \text{the predicted score is 90 by our model}$$

* Simple Linear Regression

- eg. $\text{std study_hours} \rightarrow \text{exam score}$

↳ One numerical target

$$y = b_0 + b_1 x$$

* Use case

- ## * Multiple Linear Regression

- $$y = b_0 + b_1 x_1 + b_2 x_2 + b_3 x_3$$

Eg. study-hr
attendance
internet-usage } Exam score

Simple	Multiple
One feature	Multiple feature
Easier	More realistic
$y = b_0 + b_1 x$	$y = b_0 + b_1 x_1 + b_2 x_2 + b_3 x_3$

Polynomial Regression

- Regression technique that models a curved relationship by adding powers of the features.

Instead of, $y = b_0 + b_1x$

we can do $y = b_0 + b_1x + b_2x^2$

or, $y = b_0 + b_1x + b_2x^2 + b_3x^3$

Example, $y = 2 + 3x + 2x^2$

where, $x = 2$

$$y = 2 + 3 \times 2 + 2 \times 2^2$$

$$= 2 + 6 + 8$$

$$= 16$$

x^2 term allows the model to represent a curve.

- Use polynomial regression when features and target appears curved rather than straight.

- Even though output can be curved, the model is still linear with respect to its coefficients.

For Eg. $y = b_0 + b_1x + b_2x^2$

x is non linear but b_0, b_1, b_2 are still linear.

Error

Difference between the actual value and the value predicted by the model.

$$\text{Error} = \text{Actual} - \text{Predicted}$$

Eg. actual score = 80

Predicted = 75

Error = $80 - 75 = 5$ - marks away from actual score.

Why do we need Errors?

- Because a prediction by itself does not tell us whether the model is good.

Actual Score	Model A	Model B	
80	79	50	— Model A better
65	62	70	— Model A better

Positive and Negative Error

Situation	Error	
Prediction too low	Positive	Underestimation
Prediction too high	Negative	Overestimation
Prediction exact	0	

Residual

Residual is the difference between actual value and predicted value in a regression model.

$$\text{Residual} = \text{Actual} - \text{Predicted}$$

- Residual are used to measure prediction mistakes
- compare models
- understand whether a regression model is appropriate.
- calculate regression evaluation metrics like MAE, MSE, RMSE, R^2 .

Problem with adding Errors

Student	Actual	Predicted	Error	
A	80	75	5	- Adding errors = $5 - 5 + 5 - 5$ $= 0$ - +ve & -ve errors can cancel out so, we use <u>Squared Error</u>
B	60	65	-5	
C	90	85	5	
D	70	75	-5	

Squared Error

- square of the prediction error.

$$\text{Squared Error} = (\text{Actual} - \text{Predicted})^2$$

We use squared error

- to prevent cancellation
- to penalize large mistakes more heavily

Eg. Error A = 2

Squared = 4

Error B = 10

Squared = 100

→ 5 times larger

→ 25 times larger

Squared error is heavily used in machine learning regression.

It forms

$$\text{MSE (Mean Squared Error)} = \frac{1}{n} \sum (\text{Actual} - \text{Predicted})^2$$

$$= \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

MSE tells us how far our model's predictions are from actual values while giving extra importance to large mistakes.

Commonly used in - Linear regression, multiple linear regression, polynomial regression, neural-network regression, model evaluations etc.

Best Fit Line

- Line that provides the best overall predictions for the given data according to the chosen fitting criterion

Example

Actual	Line A	Line B
50	48	40
60	61	70
70	69	60
80	82	90
MSE	2.5	12.5

30

85

80

75

70

65

60

55

50

45

40

35

X

Y

1 2 3 4 5 6 7

Line A is better & best fit

it also produces the smallest ^{overall} ~~least~~ squared error.

Loss / Cost Function

loss function - measures how bad our model's prediction is

eg. actual = 80

prediction = 75

error = $80 - 75 = 5$

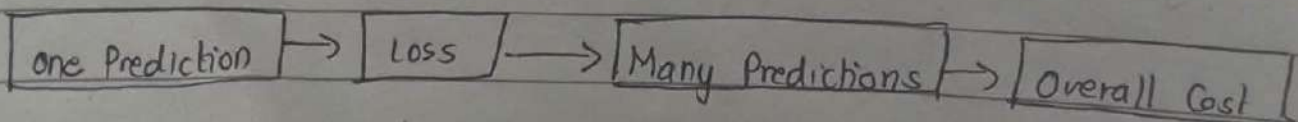
Squared loss = $5^2 = 25$

Cost

summarizes the loss over many training examples.

$$\text{Cost} = \frac{1}{n} \sum (Actual - Predicted)^2$$

$$m) \text{ Cost function} = \frac{1}{2n} \sum_{i=1}^n (x_i b - y_i)^2$$



Ordinary Least Squared (OLS)

✓ method in a linear regression to find the values of the slope and intercept that produces the smallest possible sum of squared residuals.

✓ OLS finds the line that minimizes the total squared predⁿ errors.

Our eqⁿ $\hat{y} = b_0 + b_1 x$ where $b_0 = \text{intercept}$

$b_1 = \text{slope}$

There are many possible lines. e.g. $y = 10x + 5x$

$$y = 20 + 8x$$

$$y = 30 + 10x$$

OLS helps us to identify which slope & intercept produces the smallest total squared error:

$$b_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

Ex. study-hrs(x)	exam score(y)	$x - \bar{x}$	$(y - \bar{y})$	$(x - \bar{x})^2$	$(x - \bar{x})(y - \bar{y})$
1	40	-2	-20	4	40
2	50	-1	-10	1	10
3	60	0	0	0	0
4	70	1	10	1	10
5	80	2	20	4	40
				10	100

$$\bar{x} = 3$$

$$\bar{y} = 60$$

$$b_1 = \frac{100}{10} = 10 \rightarrow \text{when } x = 1$$

$$\hat{y} = 30 + 10x$$

$$= 30 + 10 = 40 - \text{exact}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

$$= 60 - 10 \times 3$$

$$= 30$$

when $x = 4$

$$\hat{y} = 30 + 10 \times 4$$

$$= 70 - \text{exact}$$

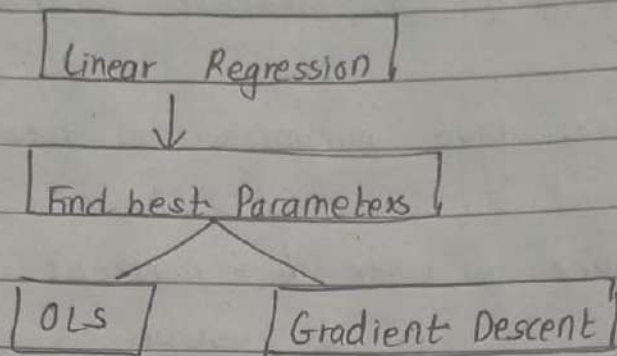
$$\hat{y} = 30 + 10x$$

$$\hat{y} = b_0 + b_1 x$$

Our line perfectly matches the dataset.

Gradient Descent

- uses an iterative optimization process to gradually adjust parameters to reduce the loss.



Gradient descent tries a set of parameters, checks how bad the predictions are, and then adjusts the parameters to make the error smaller.

Gradient - tells us the direction in which the function changes most strongly.

Steps

1. Initialize the value

$$x := x - \alpha \frac{df(x)}{d(x)}$$

Repeat step 1 and 2

until you get optimal value.

$$\alpha (\alpha) = \text{Hyperparam}$$

Hyper parameter = Learning rate

Learning rate (α)

Parameter that controls how large a step Gradient Descent takes when updating model parameter

Iteration - one cycle in which Gradient Descent calculates the necessary update and change the model parameters.

Why do we split data?

Suppose students = 1000

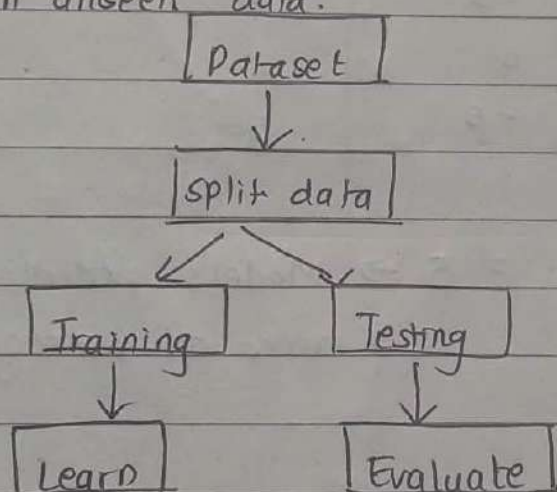
we can train model using all 1000 data

But, How do we know whether model works on students it has never seen before?

That's why we split data.

Train/Test Split

- divides the dataset into a training set used to learn the model and test set used to evaluate how well the trained model performs on unseen data.



To evaluate the model we use evaluation metrics.

1. Mean Absolute Error (MAE)
2. Mean Squared Error (MSE)
3. Root Mean Squared Error (RMSE)
4. R^2
5. Adjusted R^2

1. Mean Absolute Error (MAE)

- average absolute difference between actual values and predicted values.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

$$MAE = \frac{1}{n} \sum |Actual - Predicted|$$

Example.

actual (y_i)	Predicted ($\hat{y}_i$)	Error	Absolute Error
80	75	5	5
60	65	-5	5
90	85	5	5
70	75	-5	5

$$MAE = \frac{5+5+5+5}{4} = 5 \Rightarrow \text{Model's predictions are off by 5 marks on average in absolute terms}$$

2. Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$= \frac{1}{n} (25)^2 = \frac{625}{25} = 25$$

3. Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

$$RMSE = \sqrt{MSE}$$

$$= \sqrt{25}$$

$$= 5$$

4. R^2

- measures how much of the variation in the target variable is explained by the regression model relative to a baseline that always predicts the mean target value.

- commonly called coefficient of determination

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

$\hat{y}$ = predicted

$\bar{y}$ = mean / average

$RE = 0$ = no error

$R^2 = 1 \uparrow$ Good performance

$R^2 = 0 \downarrow$ Low performance

$R^2 = -1 \downarrow$ worst performance

$$\text{Adjusted } R^2 = 1 - \frac{(1 - R^2)(n-1)}{(n-f-1)}$$

n = no. of rows

f = features

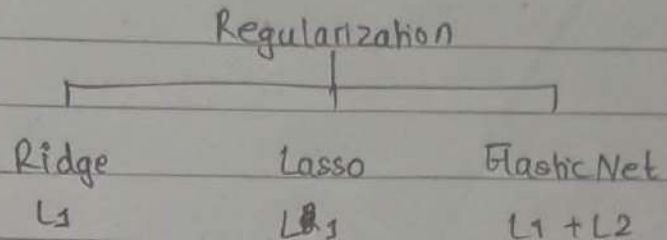
Regularization

Technique that prevents the model from becoming unnecessarily complicated.

- To solve the issue of overfitting.

Types

- L1 Regularization
- L2 Regularization



$$E = \sum_{i=1}^n (y_i - x_i m - b)^2$$

L1 Lasso

- Linear regression technique that adds an L1 penalty, proportional to the sum of absolute value of coefficients, to OLS objective function.

$$E = \sum_{i=1}^n (y_i - x_i m - (\bar{y} - m\bar{x}))^2 + \alpha |m|$$

L2 Ridge

$$E = \sum_{i=1}^n (y_i - x_i m - (\bar{y} - m\bar{x}))^2 + \alpha (m)^2$$

Elastic Net

- Regularized linear regression method that combines L1 & L2 penalties in the same objective function.